

Time Series Analysis

Chapter 4 — Non-stationary time series

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Chapter 1

Non-stationary time series

Introduction

Most real-world time series are *not* stationary: GDP grows, prices drift, populations expand, atmospheric CO₂ accumulates. Applying the tools of Chapter ?? (ACF, ARMA) to a raw non-stationary series gives misleading results — the famous *spurious regression* phenomenon. This chapter formalises non-stationarity, introduces the algebraic operators (backshift, difference) and the unit-root tests (ADF, KPSS) needed to detect it, and culminates in the *ARIMA* and *SARIMA* families that combine differencing with the ARMA structure of Chapter ??.

1.1 Limits of the stationarity assumption

Warning – Spurious regression

If X_t and Y_t are two *independent* random walks and one fits the linear regression $Y_t = \beta_0 + \beta_1 X_t + u_t$ by OLS:

- the standard t -statistic for $\hat{\beta}_1$ rejects the null $\beta_1 = 0$ at any level with probability tending to *one* as the sample size grows;
- the R^2 stays large (often > 0.5);

even though the two series are *independent*. Granger & Newbold (1974) called this the *spurious regression* effect; it is caused by the non-stationarity of X and Y .

Remark – Practical consequence

Before applying any classical inference (regression, ARMA, t -test), *check stationarity*. If the series is non-stationary, transform it first (typically by differencing).

1.1.1 Trend stationary vs. difference stationary

Definition 1 – TS and DS processes

- *Trend stationary* (TS): $X_t = f(t) + \varepsilon_t$ where f is deterministic and ε_t is stationary. Shocks have *transient* effects.
- *Difference stationary* (DS): ΔX_t is stationary, but X_t is not. The series has a *unit root*; shocks have *permanent* effects.

Remark

TS series are made stationary by *detrending*; DS series, by *differencing*. Visual inspection alone cannot reliably distinguish the two — a slowly trending DS process looks identical to a TS one. We use *unit-root tests* (Section 1.3).

1.2 Backshift and difference operators

Definition 2 – Backshift operator

$BX_t = X_{t-1}$, and $B^k X_t = X_{t-k}$. For polynomials $\phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p$ and $\theta(z) = 1 + \theta_1 z + \dots + \theta_q z^q$, the AR(p) and MA(q) processes are written $\phi(B)X_t = \varepsilon_t$ and $X_t = \theta(B)\varepsilon_t$.

Proposition 3 – Stationarity through the AR polynomial

The AR(p) process $\phi(B)X_t = \varepsilon_t$ is (weakly) stationary if and only if every root of $\phi(z) = 0$ lies *outside* the closed unit disk.

Example 4 – AR(1)

$\phi(z) = 1 - \phi z$ has root $z = 1/\phi$. $|1/\phi| > 1 \Leftrightarrow |\phi| < 1$. We recover the familiar AR(1) stationarity condition.

Definition 5 – Difference operator

$\Delta = 1 - B$, so $\Delta X_t = X_t - X_{t-1}$. $\Delta^d = (1 - B)^d$ is the d -th difference. The seasonal difference of period s is $\Delta_s = 1 - B^s$.

Example 6 – Higher-order differencing

$\Delta^2 X_t = (1 - B)^2 X_t = X_t - 2X_{t-1} + X_{t-2}$. A quadratic trend $X_t = a + bt + ct^2$ is reduced to a constant by Δ^2 : $\Delta^2 X_t = 2c$, which is stationary.

Definition 7 – Integration order $I(d)$

$X_t \sim I(d)$ if $\Delta^d X_t$ is stationary but $\Delta^{d-1} X_t$ is not. White noise is $I(0)$; a random walk is $I(1)$; a random walk with a stochastic trend is often $I(2)$.

1.3 Unit-root tests

1.3.1 Augmented Dickey–Fuller (ADF) test

Definition 8 – ADF regression

Starting from an AR(1) $X_t = \rho X_{t-1} + \varepsilon_t$, substitute $\rho = 1 + \gamma$ and subtract X_{t-1} :

$$\Delta X_t = \gamma X_{t-1} + \varepsilon_t,$$

augmented with p lags of ΔX_t to whiten the error:

$$\Delta X_t = \gamma X_{t-1} + \sum_{i=1}^p \alpha_i \Delta X_{t-i} + \varepsilon_t.$$

Theorem 9 – ADF test

- $H_0 : \gamma = 0$ — unit root, X_t is non-stationary.
- $H_1 : \gamma < 0$ — X_t is stationary.

The t -statistic of $\hat{\gamma}$ *does not* follow a Student's t under H_0 ; its asymptotic distribution is the *Dickey–Fuller distribution*, with 5% critical value ≈ -2.86 for the constant-only specification (and ≈ -3.41 with constant and trend).

Remark – Three flavours

- *No constant, no trend* (rare): X_t should fluctuate around zero.
- *Constant only* (most common): X_t should fluctuate around a non-zero level.
- *Constant and trend*: X_t has a deterministic linear trend.

The choice affects the critical values; the Python function `adfuller` accepts a `regression` argument.

1.3.2 KPSS test**Definition 10 – KPSS test, Kwiatkowski et al. 1992**

H_0 : stationarity (around a level or a trend), H_1 : unit-root non-stationarity.

The test statistic is built from the partial sums of OLS residuals versus a long-run variance estimator. 5% critical values: 0.463 for level stationarity, 0.146 for trend stationarity. Reject H_0 if the statistic *exceeds* the critical value.

Remark – Combining ADF and KPSS

The two tests have *opposite* null hypotheses, which clarifies ambiguous cases:

	ADF rejects	ADF does not reject
KPSS rejects	contradiction	non-stationary
KPSS does not reject	stationary	inconclusive (small T)

The contradiction case typically arises when there is a structural break.

1.3.3 Phillips–Perron test (mention)

Remark

The *Phillips–Perron* (PP) test is an alternative to ADF that adjusts the test statistic for heteroscedasticity and autocorrelation non-parametrically (instead of via lagged differences). Same null hypothesis as ADF, similar critical values.

1.4 ARIMA(p, d, q)

Definition 11 – ARIMA model

$X_t \sim \text{ARIMA}(p, d, q)$ if its d -th difference is a stationary $\text{ARMA}(p, q)$:

$$\phi(B)(1-B)^d X_t = \theta(B)\varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2).$$

Remark – Practical recipe

1. Plot X_t ; identify visual non-stationarity.
2. Test (ADF + KPSS); difference until stationary; choose d .
3. ACF/PACF of $\Delta^d X_t \Rightarrow$ candidate (p, q) .
4. MLE estimate of the $\text{ARMA}(p, q)$ coefficients.
5. Ljung–Box on residuals; if not white, increase p or q .
6. Forecast, integrate back.

Example 12 – ARIMA(1, 1, 1)

$\phi(B) = 1 - \phi B$, $\theta(B) = 1 + \theta B$, $d = 1$:

$$(1 - \phi B)(1 - B)X_t = (1 + \theta B)\varepsilon_t.$$

With $W_t = \Delta X_t$: $W_t = \phi W_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}$, i.e. $W_t \sim \text{ARMA}(1, 1)$. Starting from a known initial value X_0 , integrate back: $X_t = X_0 + \sum_{s=1}^t W_s$.

1.5 Seasonal ARIMA (SARIMA)

When the series exhibits seasonality of period s , classical differencing $(1-B)^d$ alone is not enough — we also apply seasonal differencing $(1-B^s)^D$ and seasonal ARMA polynomials.

Definition 13 – SARIMA(p, d, q)(P, D, Q)_s

$$\phi(B)\Phi(B^s)(1-B)^d(1-B^s)^D X_t = \theta(B)\Theta(B^s)\varepsilon_t,$$

where $\Phi(z) = 1 - \Phi_1 z - \dots - \Phi_P z^P$ and $\Theta(z) = 1 + \Theta_1 z + \dots + \Theta_Q z^Q$ are the *seasonal* AR and MA polynomials.

Example 14 – SARIMA(1, 1, 1)(1, 1, 1)₁₂

$$(1 - \phi B)(1 - \Phi B^{12})(1 - B)(1 - B^{12})X_t = (1 + \theta B)(1 + \Theta B^{12})\varepsilon_t.$$

This model is the workhorse for monthly time series with both short-term persistence and yearly seasonality (sales, climate, energy demand).

Remark – Identification

After the differencing step, look at the ACF/PACF of $(1 - B)^d(1 - B^s)^D X_t$:

- significant spikes at lags $s, 2s, 3s, \dots$ in the ACF $\Rightarrow$ seasonal MA component (order Q);
- spikes at $s, 2s, \dots$ in the PACF $\Rightarrow$ seasonal AR component (order P);
- spikes at small lags $\Rightarrow$ ordinary p, q .

Summary

Key takeaways.

- Most real series are non-stationary; *spurious regression* can fool naive applications of OLS or ARMA.
- Distinguish *trend stationary* (detrend) from *difference stationary* (difference) using the ADF and KPSS tests.
- The backshift B and difference Δ operators give a compact algebra for non-stationary models; integration order $I(d)$ counts the number of differencings needed.
- $ARIMA(p, d, q) =$ ARMA on the d -th difference; *SARIMA* adds seasonal AR, MA and differencing.
- In Python, `statsmodels.tsa.stattools.adfuller` and `kpss`, plus `SARIMAX`, cover the whole workflow.